

Question				Marking details	Marks Available																					
					Total	AO1	AO2	AO3	Maths	Prac																
11.	(a)			award (1) for either of following a molecule that partially dissociates to release/donate H ⁺ a molecule that releases H ⁺ in a reversible reaction	1	1																				
	(b)			award (1) each for calculating K _a or pK _a or pH for two acids <table border="1"><tr><td></td><td>K_a / mol dm⁻³</td><td>pK_a</td><td>pH</td></tr><tr><td>X = H</td><td>1.32 × 10⁻⁵</td><td>4.88</td><td>2.59</td></tr><tr><td>X = Cl</td><td>1.48 × 10⁻³</td><td>2.83</td><td>1.57</td></tr><tr><td>X = OH</td><td>1.38 × 10⁻⁴</td><td>3.86</td><td>2.08</td></tr></table> X = H < X = OH < X = Cl (1) ecf possible from calculated values		K _a / mol dm ⁻³	pK _a	pH	X = H	1.32 × 10 ⁻⁵	4.88	2.59	X = Cl	1.48 × 10 ⁻³	2.83	1.57	X = OH	1.38 × 10 ⁻⁴	3.86	2.08	3		1	1	2	
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	(c)	(i)		[H ⁺] = $\frac{K_w}{[OH^-]}$ = 1.25 × 10 ⁻¹⁴ mol dm ⁻³ (1) pH = 13.9 (1) ecf possible from incorrect [H ⁺]	2		2		2																	
		(ii)		concentration of salt = concentration of acid so pH = pK _a (1) pH = 4.74 (1)	2		1	1	2																	

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		(iii)		$\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$ (1) award (1) each for any two of following $\text{CH}_3\text{COONa} \rightarrow \text{CH}_3\text{COO}^- + \text{Na}^+$ - addition of small amount of H^+ - this shifts equilibrium to left to remove H^+ - addition of small amount of OH^- - this removes H^+ and shifts equilibrium to right to produce more H^+	3	3				
					11	4	4	3	6	0